

Exercise 1.

$$(a) \int_0^\infty x^3 e^{-2x} dx = \int_0^\infty x^{4-1} e^{-\frac{x}{1/2}} dx = \left(\frac{1}{2}\right)^4 \Gamma(4) = \frac{1}{16} 3! = \frac{1}{16} \times 6 = \frac{3}{8}$$

$$(b) \int_0^\infty \sqrt{x} e^{-x/2} dx = \int_0^\infty x^{1/2} e^{-x/2} dx = \int_0^\infty x^{3/2-1} e^{-x/2} dx = (2)^{3/2} \Gamma\left(\frac{3}{2}\right) = 2\sqrt{2} \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \sqrt{2\pi}$$

(c) Gaussian integral

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

$$\int_{-\infty}^{\infty} e^{-2x^2} dx$$

$$u = \sqrt{2}x$$

$$du = \sqrt{2}dx$$

$$\frac{1}{2} \int_{-\infty}^{\infty} e^{-u^2} du$$

$$\frac{1}{\sqrt{2}} \times \sqrt{\pi} = \sqrt{\frac{\pi}{2}}$$

(d)

$$\sum_{x=4}^{\infty} \binom{x-1}{3} \left(\frac{2}{3}\right)^{x-4}$$

$$x - 1 = n + 3$$

$$\sum_{n=0}^{\infty} \binom{n+3}{3} \left(\frac{2}{3}\right)^n$$

$$r = \frac{2}{3}, k = 3$$

$$\sum_{n=0}^{\infty} \binom{n+k}{k} r^n = \frac{1}{(1-r)^{k+1}} \text{ for } |r| < 1$$

$$= \frac{1}{(1-\frac{2}{3})^{3+1}} \text{ for } |r| < 1$$

$$= \frac{1}{(1-\frac{2}{3})^4}$$

$$= \frac{1}{(\frac{1}{3})^4}$$

$$= \frac{1}{\frac{1}{81}}$$

$$= 81$$

$$\begin{aligned} (\mathbf{e}) &= \sum_{x=0}^n \frac{1}{x!(n-x)!} \text{ for } n > 1 \\ &= \frac{1}{n!} \sum_{x=0}^n \binom{n}{x} \\ &= \frac{2^n}{n!} \end{aligned}$$

Exercise 2.

$$f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1} 1_{(0,1)}(x)$$

$$E(X) = \int_0^1 x \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1} dx$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 x^\alpha (1-x)^{\beta-1} dx$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 x^{\alpha+1-1} (1-x)^{\beta-1} dx$$

$$\int_0^1 t^{x-1} (1-t)^{y-1} dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \times \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+\beta+1)}$$

$$= \frac{\Gamma(\alpha+\beta)\Gamma(\alpha+1)}{\Gamma(\alpha)\Gamma(\alpha+\beta+1)}$$

$$= \frac{\Gamma(\alpha+\beta)\alpha\Gamma(\alpha)}{\Gamma(\alpha)(\alpha+\beta)\Gamma(\alpha+\beta)}$$

$$\mu = \frac{\alpha}{\alpha+\beta}$$

$$Var(X) = E(X^2) - E(X)^2$$

$$E(X^2) = \int_0^1 x^2 \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1} dx$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 x^{\alpha+1} (1-x)^{\beta-1} dx$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 x^{\alpha+2-1} (1-x)^{\beta-1} dx$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \times \frac{\Gamma(\alpha+2)\Gamma(\beta)}{\Gamma(\alpha+\beta+2)}$$

$$= \frac{\Gamma(\alpha+\beta)\Gamma(\alpha+2)}{\Gamma(\alpha)\Gamma(\alpha+\beta+2)}$$

$$\begin{aligned}
&= \frac{\Gamma(\alpha+\beta)(\alpha+1)(\alpha)\Gamma(\alpha)}{\Gamma(\alpha)(\alpha+\beta+1)(\alpha+\beta)\Gamma(\alpha+\beta)} \\
&= \frac{\alpha(\alpha+1)}{(\alpha+\beta+1)(\alpha+\beta)}
\end{aligned}$$

$$\begin{aligned}
Var(X) &= \frac{\alpha(\alpha+1)}{(\alpha+\beta+1)(\alpha+\beta)} - \left(\frac{\alpha}{\alpha+\beta}\right)^2 \\
&= \frac{(\alpha^2+\alpha)}{(\alpha+\beta+1)(\alpha+\beta)} - \frac{\alpha^2}{(\alpha+\beta)^2} \\
&= \frac{(\alpha^2+\alpha)(\alpha+\beta) - \alpha^2(\alpha+\beta+1)}{(\alpha+\beta+1)(\alpha+\beta)^2} \\
&= \frac{\alpha^3 + \alpha^2 + \alpha^2\beta + \alpha\beta - \alpha^3 - \alpha^2\beta - \alpha^2}{(\alpha+\beta+1)(\alpha+\beta)^2} \\
&= \frac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2}
\end{aligned}$$

Exercise 3.

$X \sim \text{Poisson}(\lambda)$

$$P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}, x = 0, 1, 2, 3, \dots$$

$$\begin{aligned}
E((X)_r) &= \sum_{x=r}^n x_r e^{-\lambda} \frac{\lambda^x}{x!} \\
&= \sum_{x=r}^n \frac{x!}{(x-r)!} e^{-\lambda} \frac{\lambda^x}{x!} \\
&= e^{-\lambda} \sum_{x=r}^n \frac{\lambda^x}{(x-r)!}
\end{aligned}$$

$$k = x - r$$

$$\begin{aligned}
&= e^{-\lambda} \sum_{k=0}^n \frac{\lambda^{k+r}}{k!} \\
&= e^{-\lambda} \sum_{k=0}^n \frac{\lambda^k \lambda^r}{k!} \\
&= e^{-\lambda} \lambda^r \sum_{k=0}^n \frac{\lambda^k}{k!}
\end{aligned}$$

Maclaurian Series Expansion

$$\sum_{n=0}^{\infty} \frac{1}{n!} x^n = e^x$$

$$= e^{-\lambda} \lambda^r e^{\lambda}$$

$$E((X)_r) = \lambda^r$$

Exercise 4.

$$Z \sim N(0, 1)$$

$$\text{pdf: } = \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} \text{ for } -\infty < x < \infty$$

$$\text{(a) } E(Z) = \int_{-\infty}^{\infty} z \frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi}} dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z e^{-\frac{z^2}{2}} dz$$

$$= \frac{1}{\sqrt{2\pi}} \times -2e^{-\frac{1}{2}z^2} \Big|_{-\infty}^{\infty}$$

$$= 0$$

$$E(Z^2) = \int_{-\infty}^{\infty} z^2 \frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi}} dz$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} z^2 e^{-\frac{1}{2}z^2} dz$$

$$u = z, dv = z e^{-\frac{1}{2}z^2} dz$$

$$du = dz, v = -e^{-\frac{1}{2}z^2}$$

$$= \frac{1}{\sqrt{2\pi}} [-z e^{-\frac{1}{2}z^2} + \int_0^{\infty} e^{-\frac{1}{2}z^2} dz]$$

$$= \frac{1}{\sqrt{2\pi}} [-z e^{-\frac{1}{2}z^2} - 2e^{-\frac{1}{2}z^2} \Big|_0^{\infty}]$$

$$= 1$$

$$E(Z^{2m-1}) = \int_{-\infty}^{\infty} z^{2m-1} \frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi}} dz$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} z^{2m-1} e^{-\frac{1}{2}z^2} dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} z^{2m-2} 2z e^{-\frac{1}{2}z^2} dz$$

$$u = z^{2m-2}, dv = z e^{-\frac{1}{2}z^2}$$

$$du = (2m-2)z^{2m-3}, v = -e^{-\frac{1}{2}z^2}$$

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi}} \left[-z^{2m-2} e^{-\frac{1}{2}z^2} + (2m-2) \int_0^\infty z^{2m-3} e^{-\frac{1}{2}z^2} dz \right] \\
&= \frac{1}{\sqrt{2\pi}} \left[-z^{2m-2} e^{-\frac{1}{2}z^2} + (2m-2) E(Z^{2m-3}) \right] \\
&= 0 + \frac{1}{\sqrt{2\pi}} (2m-2) E(Z^{2m-3}) \\
&E(Z^{2m-1}) = (2m-2)(2m-4)(2m-6)\dots(4)(2)E(Z) \\
&E(Z^{2m-1}) = 0
\end{aligned}$$

$$\begin{aligned}
E(Z^{2m}) &= \int_{-\infty}^\infty z^{2m} \frac{e^{-\frac{1}{2}z^2}}{\sqrt{2\pi}} dz \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty z^{2m} e^{-\frac{1}{2}z^2} dz \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty z^{2m-1} z e^{-\frac{1}{2}z^2} dz
\end{aligned}$$

$$u = z^{2m-1}, dv = z e^{-\frac{1}{2}z^2}$$

$$du = (2m-1)z^{2m-2}, v = -ze^{-\frac{1}{2}z^2}$$

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi}} \left[z^{2m-1} - ze^{-\frac{z^2}{2}} + \int_{-\infty}^\infty ze^{-\frac{1}{2}z^2} (2m-1)z^{2m-2} dz \right] \\
&= \frac{1}{\sqrt{2\pi}} \left[z^{2m-1} - ze^{-\frac{z^2}{2}} + (2m-1) \int_{-\infty}^\infty ze^{-\frac{1}{2}z^2} z^{2m-2} dz \right] \\
&= \frac{1}{\sqrt{2\pi}} \left[0 + (2m-1) \int_{-\infty}^\infty e^{-\frac{1}{2}z^2} z^{2m-1} dz \right] \\
&= \frac{(2m-1)}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{-\frac{1}{2}z^2} z^{2m-1} dz \\
&= \frac{1}{\sqrt{2\pi}} (2m-1) E(Z^{2m-2}) \\
&= (2m-1)(2m-3)(2m-5)\dots(3)E(Z^2) \\
&= (2m-1)(2m-3)(2m-5)\dots(3)(1) \\
&= (2m-1)!!
\end{aligned}$$

Properties of Double Factorial

$$n!! = \prod_{k=1}^{\frac{n+1}{2}} (2k-1) = n(n-2)(n-4)\dots 3 * 1$$

$$n!! = \frac{n!}{(n-1)!!} = \frac{(n+1)!}{(n+1)!!}$$

for an even non-negative integer $n=2k$ with $k \geq 0$

$$2k!! = 2^k k!$$

$$(2k-1)!! = \frac{(2k)!}{2^k k!} = \frac{(2k-1)!}{2^{k-1}(k-1)!}$$

$$= (2m-1)!! = \frac{(2m)!!}{2^m m!}$$

$$(b) \int_{-\infty}^{\infty} \cos(\theta z) \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$$

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

$$\begin{aligned} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{e^{i\theta z} + e^{-i\theta z}}{2} e^{-\frac{z^2}{2}} dz \\ &= \frac{1}{2\sqrt{2\pi}} \int_{-\infty}^{\infty} (e^{i\theta z} + e^{-i\theta z}) e^{-\frac{z^2}{2}} dz \\ &= \frac{1}{2\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i\theta z - \frac{z^2}{2}} + e^{i\theta z + \frac{z^2}{2}} dz \\ &= \frac{1}{2\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{1}{2}(2i\theta z - z^2)} + e^{\frac{1}{2}(2i\theta z + z^2)} dz \\ &= \frac{1}{2\sqrt{2\pi}} \left[\int_{-\infty}^{\infty} e^{\frac{1}{2}(2i\theta z - z^2)} + \int_{-\infty}^{\infty} e^{\frac{1}{2}(2i\theta z + z^2)} dz \right] \end{aligned}$$

both integrals are equal to each other since the normal distribution is symmetric.

$$\begin{aligned} &= \frac{2}{2\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{1}{2}((z-i\theta)^2 - \theta^2)} \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-i\theta)^2} e^{-\frac{\theta^2}{2}} dz \\ &= \frac{e^{-\frac{\theta^2}{2}}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-i\theta)^2} \end{aligned}$$

$$u = z - i\theta$$

$$du = dz$$

$$\begin{aligned} &= e^{-\frac{\theta^2}{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2} dx \text{ (the integral is equal to 1 because its the pdf integrated)} \\ &= e^{-\frac{\theta^2}{2}} \times 1 \\ &= e^{-\frac{\theta^2}{2}} \end{aligned}$$

Exercise 5.

$$Y = e^{\delta T}$$

$$T\delta = \ln(Y)$$

$$T = \frac{\ln(Y)}{\delta}$$

for $y > 1$ (i.e. $e^{\delta T} \geq e^{\delta 0} = 1$)

$$F_Y(y) = P(Y \leq y) = P(e^{\delta T} \leq y)$$

$$P(e^{\delta T} < y) = P(T \leq \frac{\ln(y)}{\delta})$$

$$P(T \leq t) = 1 - e^{-ct}$$

$$F_Y(T \leq \frac{\ln(y)}{\delta}) = 1 - e^{-c \frac{\ln(y)}{\delta}}$$

$$F_Y = 1 - y^{-\frac{c}{\delta}}$$

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

$$= \frac{d}{dy} (1 - y)^{-\frac{c}{\delta}}$$

$$f_Y(y) = \frac{c}{\delta} y^{-\frac{c}{\delta}-1}$$

This resembles a Pareto distribution, where $\alpha = \frac{c}{\delta}$ and $\lambda = 1$

Exercise 6.

(a) $Y \sim N(\mu, \sigma^2)$

$$Y \sim N(\frac{5}{9}(88 - 32), (\frac{5}{9}2)^2)$$

$$Y \sim N(31.1111, 1.2346)$$

(b) $P(X > 93)$

$$z = \frac{x-\mu}{\sigma} = \frac{93-88}{2} = 2.5$$

$$P(X > 93) = P(Z > 2.5)$$

$$Z = 1 - 0.9938$$

$$P(X > 93) = 0.0062$$

(c) $P(Y < 30)$

$$z = \frac{30-31.1111}{1.1111} = \frac{-1.1111}{1.1111} = -1$$

$$P(Y < 30) = P(Z < -1)$$

$$Z = 0.1587$$

$$P(Y < 30) = 0.1587$$

Exercise 7.

- hypotenuse X , continuous random variable having PDF $f(x) = 3x^2 1_{(0,1)}(x)$

(a) $X^2 = L^2 + L^2$

$$X^2 = 2L^2$$

$$L^2 = \frac{X^2}{2}$$

$$L = \frac{X}{\sqrt{2}}$$

Area of Triangle is:

$$Y = \frac{1}{2}L^2$$

$$Y = \frac{1}{2}\left(\frac{X}{\sqrt{2}}\right)^2$$

$$Y = \frac{1}{2} \times \frac{X^2}{2}$$

$$Y = \frac{X^2}{4}$$

$$X^2 = 4Y$$

$$X = 2\sqrt{Y}$$

$$\frac{dX}{dY} = \frac{d}{dY}(2\sqrt{y}) = \frac{1}{\sqrt{y}}$$

$$f_Y(y) = f_X(x) \left| \frac{dX}{dY} \right| = f_x(2\sqrt{2}) \times \frac{1}{\sqrt{y}}$$

$$= 3(2\sqrt{y})^2 \times \frac{1}{\sqrt{y}}$$

$$= 3 \times 4y \times \frac{1}{\sqrt{y}}$$

$$f_Y(y) = 12\sqrt{y}$$

Supp of X is $(0, 1)$

$$(0, \frac{X^2}{4}, X = 1)$$

$$(0, \frac{1}{4})$$

Supp of Y: $0 < Y < \frac{1}{4}$

$$\begin{aligned}
 \text{(b)} \quad P(Y < \frac{1}{16}) &= \int_0^{\frac{1}{16}} 12\sqrt{y} \, dy \\
 &= \frac{12}{3} \times 2y^{3/2} \Big|_0^{\frac{1}{16}} \\
 &= \frac{24}{3} y^{3/2} \Big|_0^{\frac{1}{16}} \\
 &= 8y^{3/2} \Big|_0^{\frac{1}{16}} \\
 &= 8\left(\frac{1}{16}\right)^{3/2} \\
 &= \frac{1}{8}
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad E[X] &= \int_0^{\frac{1}{4}} y 12\sqrt{y} \, dy \\
 &= \int_0^{\frac{1}{4}} 12y^{3/2} \, dy \\
 &= \frac{12 \times 2}{5} y^{5/2} \Big|_0^{\frac{1}{4}} \\
 &= \frac{24}{5} y^{\frac{5}{2}} \Big|_0^{\frac{1}{4}} \\
 &= \frac{24}{5} \left(\frac{1}{4}\right)^{5/2} \\
 &= \frac{24}{5} \times \frac{1}{32} \\
 &= \frac{3}{20}
 \end{aligned}$$

Exercise 8.

$$F_Y(y) = P(Y \leq y) = P(aX + b \leq y)$$

$$aX + b \leq y$$

$$aX \leq y - b$$

$$X \geq \frac{y-b}{a} \quad (\text{since } a < 0)$$

$$F_Y(y) = P(X \geq \frac{y-b}{a})$$

$$P(X \geq \frac{y-b}{a}) = P\left(\frac{x-\mu}{\sigma} \geq \frac{\frac{y-b}{a}-\mu}{\sigma}\right) = P\left(Z \geq \frac{\frac{y-b}{a}-\mu}{\sigma}\right) = P\left(Z \geq \frac{y-(a\mu+b)}{a\sigma}\right)$$

$$F_Y(y) = 1 - \Phi\left(\frac{y-(a\mu+b)}{a\sigma}\right)$$

This CDF expression show that $Y = aX + b$ as a normal distribution with a mean $= a\mu + b$ and variance $= a^2\sigma^2$, so standard deviation $= \sqrt{a^2\sigma^2} = |a|\sigma$

Therefore, $Y \sim N(a\mu + b, a^2\sigma^2)$

Exercise 9.

z-score: $z = \frac{X-\mu}{\sigma}$

standardizes X to have the mean = 0 and variance = 1

$$M_z(\theta) = E[e^{\theta z}]$$

$$M_z(\theta) = E[e^{\theta \frac{x-\mu}{\sigma}}]$$

$$M_z(\theta) = E[e^{\frac{\theta}{\sigma}(x-\mu)}]$$

$$M_z(\theta) = E[e^{\frac{\theta}{\sigma}x} e^{-\frac{\theta}{\sigma}\mu}] = e^{-\frac{\theta\mu}{\sigma}} E[e^{\frac{\theta}{\sigma}x}]$$

$$M_z(\theta) = e^{-\frac{\theta\mu}{\sigma}} M_x\left(\frac{\theta}{\sigma}\right)$$

MGF of the z-score is: $M_z(\theta) = e^{-\frac{\theta\mu}{\sigma}} M_x\left(\frac{\theta}{\sigma}\right)$

The domain of the MGF of x , $M_x(\theta)$, is given as $A < \theta < B$, where $A < 0$ (possible $-\infty$) and $B > 0$ (possible $+\infty$)

$$A < \frac{\theta}{\sigma} < B$$

$$\sigma A < \theta < \sigma B$$

Because $A < 0$, would should consider A as $|A| = -A$ when considering the domain. So, the domain of the MGF of z is: $-\sigma A < \theta < \sigma B$

Exercise 10.

PDF of $Z \sim N(0, 1)$:

$$\varphi(z) = \frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi}}$$

$$\varphi'(z) = -\frac{2z}{z} \frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi}}$$

$$= -\frac{ze^{-\frac{z^2}{2}}}{\sqrt{2\pi}}$$

$$\varphi''(x) = \frac{-e^{-\frac{z^2}{2}} + z^2 e^{-\frac{z^2}{2}}}{\sqrt{2\pi}}$$

$$= \frac{e^{-\frac{z^2}{2}}(z^2 - 1)}{\sqrt{2\pi}}$$

$$0 = \frac{e^{-\frac{z^2}{2}}(z^2-1)}{\sqrt{2\pi}}$$

$$0 = e^{-\frac{z^2}{2}}(z^2 - 1)$$

$$0 = z^2 - 1$$

$$z^2 = 1$$

$$z = \pm 1$$

If $Z = N(0, 1)$, then $X = \sigma Z - \mu - N(\mu, \sigma^2)$

The inflection points of the standard distribution Z are at $z = \pm 1$. Since $X = \sigma z + \mu$ the inflection points of Z can be transformed.

- $Z = 1, x = \sigma(1) + \mu = \mu + \sigma$
- $Z = -1, x = \sigma(-1) + \mu = \mu - \sigma$

Therefore, the inflection points of $f_X(x)$, the PDF of $X - N(\mu, \sigma^2)$, occur at $x = \mu \pm \sigma$

Exercise 11.

$X - Exp$

$$f_X(x) = e^{-x} 1_{(0, \infty)}(x)$$

$$Y = \nu + \alpha X^{1/\beta}$$

$$Y - \nu = \alpha X^{1/\beta}$$

$$X^{1/\beta} = \frac{Y - \nu}{\alpha}$$

$$X = \left(\frac{Y - \nu}{\alpha}\right)^\beta$$

$$\frac{d}{dY}(x = \left(\frac{Y - \nu}{\alpha}\right)^\beta)$$

$$\frac{dX}{dY} = \frac{\beta}{\alpha} \left(\frac{Y - \nu}{\alpha}\right)^{\beta-1}$$

$$f_Y(x) = f_X(x) \left| \frac{dX}{dY} \right|$$

$$f_Y(x) = f_X\left(\left(\frac{Y - \nu}{\alpha}\right)^\beta\right) \frac{\beta}{\alpha} \left(\frac{Y - \nu}{\alpha}\right)^{\beta-1}$$

$$f_Y(x) = e^{-\left(\frac{Y - \nu}{\alpha}\right)^\beta} 1_{(\nu, \infty)}(y) \frac{\beta}{\alpha} \left(\frac{y - \nu}{\alpha}\right)^{\beta-1}$$

$$f_Y(x) = \frac{\beta}{\alpha} \left(\frac{y-\nu}{\alpha} \right)^{\beta-1} e^{-\left(\frac{y-\nu}{\alpha} \right)^\beta} 1_{(\nu, \infty)}(y)$$

$$f_Y(x) = \frac{\beta(y-\nu)^{\beta-1}}{\alpha^\beta} e^{-\left(\frac{y-\nu}{\alpha} \right)^\beta} 1_{(\nu, \infty)}(y)$$

$$f(x) = \begin{cases} \frac{\beta(y-\nu)^{\beta-1}}{\alpha^\beta} e^{-\left(\frac{y-\nu}{\alpha} \right)^\beta} 1_{(\nu, \infty)}(y), & y > \nu \\ 0, & y \leq \nu \end{cases}$$

Exercise 12.

- Let X be a continuous random variable with a strictly increasing CDF $F(x)$
- Since $F(x)$ is strictly increasing, its inverse $F^{-1}(x)$ exists. We define U as: $U = F(X)$
- want to show $U \sim \text{Unif}(0, 1)$, which means that the CDF of U , denoted $F_U(u)$ is given by: $F_U(u) = P(U \leq u) = u$ for $0 \leq u \leq 1$
- for $u < 0$, since CDF $F(x)$ only takes values in $[0, 1]$, it follows that $P(U \leq u) = 0$ for $u < 0$
- for $u \geq 1$, the CDF $F(x)$ always satisfies $F(x) \leq 1$. Therefore, for $u \geq 1$, we have $P(U \leq u) = 1$

For $0 \leq u \leq 1$, we need to show that $P(U \leq u) = u$ $F_U(u) = P(U \leq u) = P(F(X) \leq u)$

Since $F(X)$ is a strictly increasing function, the event $F(X) \leq u$ is equal to $X \leq F^{-1}(u)$
 $P(U \leq u) = P(F(X) \leq u) = P(X \leq F^{-1}(u))$

Since F is a CDF strictly increasing $F(F^{-1}(u)) = u$

Therefore,

$$F_U(u) = P(U \leq u) = u$$

This proves that $U \sim \text{Unif}(0, 1)$ since its CDF is precisely the CDF of a uniform random variable on $[0, 1]$

If we generate a uniform random variable $U \sim \text{Unif}(0, 1)$, then the transformation: $X = F^{-1}(U)$

This will give us a random variable of X with the desired distribution described by $F(X)$

Ex: $X \sim \text{Exp}(\lambda)$

$$F(x) = 1 - e^{-\lambda x} \text{ for } x \geq 0$$

We need to find the inverse of $F(x)$. Set $F(x) = u$, where $u \in [0, 1]$:

$$u = 1 - e^{-\lambda x}$$

$$e^{\lambda x} = 1 - u$$

$$x = -\frac{1}{\lambda} \ln(1 - u)$$

$$\text{If } U \sim \text{Unif}(0, 1): X = -\frac{1}{\lambda} \ln(1 - U)$$

Exercise 13.

- $X \sim N(\mu, \sigma), e^x \sim \text{lognormal}(\mu, \sigma^2)$

- $Y \sim \text{lognormal}(\mu, \sigma^2), \ln(Y) \sim N(\mu, \sigma^2)$

(a) $\ln(Y) \sim N(1, 4), \mu = 1, \sigma^2 = 4, \sigma = 2$

$$P(Y \geq 137) = P(\ln(Y) \geq \ln(137))$$

$$z = \frac{\ln(Y) - \mu}{\sigma}$$

$$= P(Y \geq 137) = P\left(\frac{\ln(137) - 1}{2} \geq Z\right)$$

$$= P\left(\frac{4.9199 - 1}{2} \geq Z\right)$$

$$= P\left(\frac{3.9199}{2} \geq Z\right)$$

$$= P(1.9599 \geq Z)$$

$$= P(Z \leq 1.96) = 0.9750$$

$$= P(Y \geq 137) = 1 - 0.9750 = 0.025$$

$$P(7.389 \geq Y \geq 73.7) = P(\ln(7.389) \geq \ln(Y) \geq \ln(73.7))$$

$$= P(1.9999 \geq \ln(Y) \geq 4.3000)$$

$$= P(2 \geq \ln(Y) \geq 4.3)$$

$$= P\left(\frac{2-1}{2} \leq Z \leq \frac{4.3-1}{2}\right)$$

$$= P(0.5 \leq Z \leq 1.65)$$

$$= P(Z \geq 0.5) = 0.6915$$

$$' = P(Z \geq 1.65) = 0.9505$$

$$P(7.389 \leq Y \leq 73.7) = 0.9505 - 0.6915 = 0.259$$

(b) The expected value of Y is: $E(Y) = e^{\mu + \frac{\sigma^2}{2}}$

The second moment (expected value of Y^2) is: $E(Y^2) = e^{2\mu + 2\sigma^2}$

The variance $Var(Y)$ is: $Var(Y) = E(Y^2) - (E(Y))^2$

$$Var(Y) = e^{2\mu + 2\sigma^2} - (e^{\mu + \frac{\sigma^2}{2}})^2$$

$$Var(Y) = e^{2\mu + 2\sigma^2} - (e^{\mu + \frac{\sigma^2}{2}})^2$$

$$= e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2}$$

$$= e^{2\mu + \sigma^2}(e^{\sigma^2} - 1)$$

Exercise 14.

$$= \sum_{x=1}^{\infty} \sum_{y=1}^x P_{X,Y}(x, y) = 1$$

$$= \sum_{x=1}^{\infty} \sum_{y=1}^x \frac{2y}{x(x+1)} \left(\frac{1}{2}\right)^x$$

$$= \sum_{x=1}^{\infty} \frac{1}{x(x+1)} \left(\frac{1}{2}\right)^x \sum_{y=1}^x 2y$$

$$= \sum_{x=1}^{\infty} \frac{1}{x(x+1)} \left(\frac{1}{2}\right)^x \times 2 \times \frac{x(x+1)}{2}$$

$$= \sum_{x=1}^{\infty} \left(\frac{1}{2}\right)^x$$

$$= \frac{\frac{1}{2}}{1 - \frac{1}{2}}$$

$$= \frac{1/2}{1/2}$$

$$= 1$$